Fiscal Year:	FY 2026	**Task Last Updated:**	FY 07/23/2025
PI Name:	Kassemi, Mohammad Ph.D.		
Project Title:	The Zero Boil-Off Tank-NonCondensibles (ZBOT-NC) Experiment		
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Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
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Comments:	NOTE (Dec 2019): former affiliation included National Center for Space Exploration Research (NCSER), per information from J. McQuillen/GRC		
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Meyer, Michael L (NASA Langley Research Center) Kouchesfahani, Manoochehr (Michigan State University)		
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Task Description:	NASA’s projected exploration program includes a series of human and robotic expeditions to low and high earth orbit, Moon, Mars, and possibly the asteroids and moons of other planets. Integral to all phases of these space and planetary expeditions is affordable and reliable cryogenic fluid storage for use in propellant or life support systems. Without safe, efficient, and flexible cryogen storage, economically justifiable human missions may not be possible. Cryogen vaporization caused by various heat leaks into a tank from its surroundings is the main cause of storage tank self-pressurization. Without dynamic and active pressure control, self-pressurization can only be countered by continuous venting of the vapor to preserve the tank’s structural integrity. This leads to unacceptable fuel loss levels for long-duration missions and has given impetus to the development of innovative zero-boil-off or reduced-boil-off pressure control designs based on mixing of the bulk fluids with or without active cooling. Unfortunately, both tank self -pressurization and pressure control are governed by intricate and complicated dynamic interactions among forced mixing, various gravity-dependent transport mechanisms in the vapor and liquid phases, and the multifaceted condensation/evaporation processes at the interface. Consequently, effective implementation and optimization of a dynamic pressure control system for space applications cannot be accomplished empirically, especially since there is a serious scarcity of relevant microgravity data. In order to address this need, and in line with the strong recommendations provided by the National Academy of Sciences’ (NAS) National Research Council (NRC) Decadal Survey, the ZBOT series of hierarchical microgravity science experiments were proposed and initiated with the following overall goals: 1. ZBOT-1–FM (Fluid Mixing): Investigate transport and phase change phenomena associated with self-pressurization & zero-boil-off (ZBO) mixing-jet pressure control in a sealed tank, in microgravity. Lay the foundation for the development and validation of a state-of-the-art two-phase Computational Fluid Dynamics (CFD) model for storage tank cryogenic fluid management (CFM). 2. ZBOT-NC (Noncondensable effects): Investigate the flow, transport, and kinetics effects of noncondensable gas (NCG) pressurants and their impact on the interfacial condensation and evaporation processes during tank self-pressurization and pressure control in microgravity. To be accompanied also with hand-in-hand CFD model development and validation. 3. ZBOT-DP (Droplet Phase Change0: Investigate the heat and mass transport and phase change characteristics of droplet injection using a spray bar during tank pressure control and chill-down operations in microgravity, together, again, with hand-in-hand CFD model development and validation. As a first step towards achieving these goals, the ZBOT-1 experiment was performed aboard the International Space Station (ISS) in 2017. The experiment met its Complete Scientific Success Criteria and proved that such a cost-effective, subscale, simulant fluid, ISS-class science experiment can provide valuable microgravity data for: • Successful and comprehensive validation and verification of state-of-the-art CFD models that will be ultimately used to design propellant tanks for future space missions. • Gaining a better understanding of the intricate transport and phase change interactions that control tank pressurization and pressure control in microgravity. • Pointing out several serious practical implications of two-phase phenomena that may prove important to future large cryogenic storage tank operations operating in microgravity. The main goal of the present ZBOT-NC experiment is to determine the important but inadequately understood and assessed impact of non-condensable pressurant gases on the interfacial evaporation/condensation and bulk transport processes that affect tank pressurization and pressure control in microgravity. Building upon the success of ZBOT-1, the specific scientific objective in this second experimental–computational investigation will be to clarify the following phenomena: 1. The nature of the transport barrier created by the NCG in the ullage and its effect on interfacial evaporation and condensation rates in microgravity. 2.The effect of a non-condensable gas on the interfacial mass transfer kinetics, and the extent of any resistance to interfacial mass transfer caused by the gas penetrating the Knudsen layer 3. The existence, nature, and extent of a unique class of Marangoni (thermocapillary) convection in the liquid that is driven by partial pressure variations in the vapor. 4. Combined impact of the above on the global pressurization/stratification and pressure reduction/destratification rates in microgravity. In order to achieve these objectives, we plan on performing: • High accuracy measurement of the ullage pressure • High accuracy local temperature measurements at multiple locations in the tank interior and on the tank wall • Accurate determination of NCG concentrations in the ullage • Visualization of the liquid-vapor interface through white light imaging; and • Non-intrusive capture of the liquid temperature fields using state-of-the-art Quantum Dot Thermometry (QTD) diagnostics that will be developed and customized in-house. The local instrumentation/sensor data and the full-field image data will be used to validate a two-phase two-component CFD storage tank model that will be developed hand-in-hand with the experiment. This will be accomplished by building upon the solid foundation of model development and validation that was initiated and carried out quite successfully during the ZBOT-1 investigation. It is anticipated that the acquisition of the relevant microgravity data and the development of the much-needed microgravity empirical correlations for mixing and destratification in the presence of noncondensable gases, together with development, verification, and validation of the CFD models and codes, that will be ultimately used in the scale-up design of the future storage tank prototype, will all contribute significantly to reduce the uncertainty, cost, and risk associated with the development of both cryogenic upper stages and in-Space or on-surface fuel depots for future NASA planetary missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of the ZBOT microgravity investigations primarily and directly address the fundamental needs of NASA and the aerospace industry for efficient, reliable, and enabling propellant and life support storage and transfer technologies for future long-duration planetary exploration. The ZBOT experiments and the associated two phase propellant tank CFD model lay the scientific foundation that support the engineering works associated with future propellant fuel depot operations in Space, and the NASA's Artemis and Mars Mission programs. The fundamental two-phase data and scientific understanding provided by the experiment and the CFD models will also aid in the improvement of a set of diverse ground-based two-phase technologies such as condensing heat exchangers, phase separators, water-handling operations in fuel cells, and storage and transport of cryogenic and non-cryogenic consumable fluids on Earth. ZBOT-1 experiment data was used to develop and validate a state-of-the art two-phase CFD model for tank self-pressurization and pressure control through jet mixing. This model has played an important role in aiding and supporting the computational tool development efforts carried by the Evolvable Cryogenics and Cryogenic Propellant Management Portfolio programs of NASA Space Technology Mission Directorate in the last decade. The ZBOT PI and Modelling Team have also worked closely with several US Aerospace companies such as Space-X, Blue Origin, and

Lockheed under the NASA Tipping Point and Human Lander System (HLS) programs where ZBOT-1 microgravity data and CFD models were used to support and guide the following undertakings: 1. The Space-X: Propellant tank hardware development for S4 flight demonstration of a large scale cryogenic propellant transfer 2. Blue Origin: Determination of pressure collapse during engine burns for the lander hydrogen tank under different flight events and stages of the burn. 3. Lockheed: Simulation of hydrogen tank cooling using spray with submerged and unsubmerged diffuser configurations in 0 g

Task Progress:

Future refueling depot operations in space require transfer of propellants from the donor tank to the receiver tank under weightlessness conditions. For long duration depot applications, it may be advantageous to perform the transfer by pressurizing the donor tank using a gaseous pressurant rather than using a mechanical pump, which is more prone to failure. Tank pressurization can be accomplished either autogenously by using a like gas (the vapor of the liquid fuel that is being stored and transferred), or by using a noncondensable pressurant gas such as helium. However, noncondensable gas pressurization seems to be a viable option, since it can be accomplished faster and with minimal additional hardware. Unfortunately, once the noncondensable gas is present in the ullage, it can interfere with the condensation and evaporation processes at the liquid-vapor interface, and therefore have important consequences for tank self-pressurization and pressure control during subsequent storage periods. The effects of noncondensable gas on phase change, especially in relation to its impact on the condensation process in two-phase heat exchangers and phase separators, have been the subject of considerable experimental and computational research. However, the same level of scientific and engineering scrutiny has not been focused on the effects of noncondensable gas on the transport and phase change processes that take place in cryogenic propellant tanks on the ground or in microgravity. The Zero Boil-Off Tank - Noncondensable (ZBOT-NC) microgravity investigation is the first hand-in-hand science experiment and Computational Fluid Dynamics (CFD) model development effort that is devoted to elucidating the impacts of noncondensable gases on the phase change and transport processes that govern pressurization and pressure control of a volatile fluid in a sealed tank in 1g and microgravity.
Important accomplishments were made in the 2024-2025 period regarding the ZBOT-NC CFD model development. The development of a two-phase, two-component Sharp Interface SI-CFD model was completed and model simulations were used to aid the design of the ZBOT-NC Experiment and its microgravity text matrix. The model solves the Navier Stokes and continuity equations for fluid velocities and the energy and species equations for the temperature and molar concentrations, or partial pressures, of the vapor and noncondensable gas in the ullage. Numerical simulations performed by the axisymmetric SI-CFD model predicted that the noncondensable gas can impact both self-pressurization and pressure control of a propellant tank in 1g and microgravity by creating a transport barrier in the ullage that reduces interfacial condensation, as well as by instigating an interface-driven Marangoni or thermocapillary flow that alters the fluid and thermal structures in the vicinity of the ullage. In the near future, this model will be validated against the microgravity data that will be collected during the International Space Station (ISS) experiment.

The preliminary CFD simulations of test points selected from the ZBOT-NC 1g and microgravity test matrix indicated that, in microgravity, the noncondensable gas can considerably reduce the rate of self-pressurization brought about by the global heating of the tank. This impact is caused more by the Marangoni effect on the liquid side than by the transport resistance inside the ullage. The computational study also indicated that the noncondensable gas can have a considerable impact on tank depressurization by subcooled jet mixing in microgravity. The NCG effects on depressurization can be significant in microgravity. Due to the weakened buoyancy force, the subcooled jet reaches the interface with almost no delay, but, due to the simultaneous effects of transport resistance in the ullage created by the noncondensable gas and the diversion of the subcooled jet flow by the Marangoni convection, which is also instigated by the noncondensable gas, there is a significant decline in both the interfacial condensation and tank depressurization rates, compared to the pure vapor tank system.

The CFD simulations indicated a noticeable and important change in the thermal structures that evolve in the liquid domain due to the thermocapillary convection. This effect is more pronounced in microgravity. As a result, the predicted temperature fields in the liquid domain during both self-pressurization and jet mixing intervals in microgravity may be quite different for the two-component system compared to the pure system. To be able to capture the change in the temperature field, a nonintrusive laser-induced diagnostic capability based on the use of quantum dots was developed for the ZBOT-NC investigation through a collaboration with ZBOT-NC Co-Investigator, Dr. Manoochehr Koochesfahani, and his team at the Michigan State University.

The second major accomplishment of the ZBOT-NC investigation in 2024-2025 was with respect to validation of the Quantum Dot Thermometry (QDT) nonintrusive whole field diagnostic technique through direct experimentation. Although preliminary breadboard testing performed previously had yielded acceptable QDT results, there were several aspects of the tests that needed improvement and further evaluation. To this end, a series of breadboard QDT validation and verification tests were undertaken with two objectives. First, the ZBOT-NC engineering team at ZIN Technologies had developed a permanent lens mounting setup, a vacuum jacket simulator with a black anodized interior and an improved laser illumination uniformity. It was necessary to properly evaluate and assess these improvements to the optical system. Second, to determine the fidelity of the QDT diagnostics, it was imperative to perform multiple test points from the ZBOT-NC flight test matrix with sufficient experimental instrumentation (Resistance Temperature Detectors/RTDs and pressure transducer), such that the ZBOT CFD team would be able to simulate the tests under actual test conditions based on identifiable boundary and initial conditions. This would allow direct comparisons of the QDT measurement against the local sensor data and CFD predictions.

To accomplish these objectives:

First, validation of QDT measurements against local sensor measurements and CFD simulations was accomplished for a 0.8 Watts strip heater self-pressurization test. The comparison of QDT results with local sensor measurement at several RTD locations and against vertical temperature profiles and whole field temperature contours, predicted by CFD at several different instances of time during self-pressurization, produced very good agreements. In particular, the comparisons underscored the consistency between CFD predictions and QDT measurement regarding the development of vertical temperature gradients in the region below the heater, and the uniformity of temperature in the region above the heater, that is established due to natural convection mixing. Next, validation of the QDT results against local sensor measurements and CFD simulation results was also performed for a 1g tank pressure control jet mixing test that immediately followed the 0.8 Watts self-pressurization test. A subcooled jet with a velocity of 15 cm/s and a temperature that was roughly 3 degrees subcooled, relative to the initial average temperature of the tank, was used for this pressure control segment. In this case, very good agreements were again obtained between QDT measurements with

	local sensor measurements at several RTD locations and with CFD predictions of both vertical and horizontal temperature profiles and whole field temperature contours at several different instances of time during the jet mixing. Furthermore, comparison between the whole field temperature contours captured by QDT and simulated by CFD clearly showed that the spread of the jet that is caused by the interactions between the subcooled jet and the surrounding warmer, and more stagnant fluid, is three-dimensional in nature. The 2D axisymmetric model that suppresses the 3D behavior of the turbulent vortices underpredicted the true jet spread. As a result, the 2D model also underpredicted the time delay for the jet to reach the liquid vapor interface and the associated delay in the pressure drop. The 3D CFD model simulations, however, predicted the jet spread and the dynamic development of the liquid temperature contours in good agreement with the QDT results and predicted the delay in tank pressure drop that was consistent with local pressure measurements. Overall, the science team was quite satisfied with the performance of the QDT diagnostics and the two-phase CFD model in these preliminary ground-based self-pressurization and jet mixing pressure control tests, especially given the uncertainty in the initial conditions of the breadboard experiments. The QDT performance will be further optimized for the flight experiment, based on the lessons learned during the breadboard tests. This effort also underscored the value of iterating between the QDT and CFD analyses that was extremely instructive for understanding the nature of the jet flow, and demonstrated the importance of whole field thermometry in guiding the CFD model validation and improvement in the ZBOT-NC investigation.
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